№ 7.1.

а)

$$\Lambda(x) = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1}$$

$\Lambda(x) > c \Rightarrow H_1$

$\Lambda(x) < c \Rightarrow H_0$ $\qquad P_{H_0}(\Lambda(x) > c) = \alpha$

$\Lambda(x) > c \Leftrightarrow x < c$

$P_{H_0}(x < c) = c = \alpha$

$K = \{x < \alpha\}$

$$W = P_{H_1}(x < \alpha) = \int_0^\alpha \frac{e^{1-x}}{e-1}\,dx = \frac{1}{e-1}\int_0^\alpha e^{1-x}\,dx = \frac{e}{e-1}\int_0^\alpha e^{-x}\,dx = \frac{e(1-e^{-\alpha})}{e-1}$$

Ошибка первого рода: α

Ошибка второго рода: $\beta = 1 - W$

б)

$$\Lambda(x) = \prod_{i=1}^{2} \frac{p_1(x_i)}{p_0(x)} = \frac{e^{2-(x_1+x_2)}}{(e-1)^2}$$

$S < c \Rightarrow H_1$ $\qquad S > c \Rightarrow H_0$

$$k_S(S) = \begin{cases} S, & 0 \le S \le 1 \\ 2-S, & 1 \le S \le 2 \\ 0, & \text{иначе} \end{cases}$$

$P_{H_0}(S < c) = \alpha$

$$\int_0^c s\,ds = \frac{c^2}{2} = \alpha \Rightarrow c = \sqrt{2\alpha}$$

$K = \{x_1 + x_2 < \sqrt{2\alpha}$

$$W = P_{H_1}(S < \sqrt{2\alpha}) = \iint_{x_1+x_2<\sqrt{2\alpha}} \frac{e^{2-(x_1+x_2)}}{(e-1)^2}\,dx_1\,dx_2 = \frac{e^2}{(e-1)^2}\left(1 - e^{-\sqrt{2\alpha}}(1+\sqrt{2\alpha})\right)$$

с)

$$L_n = \sum_{i=1}^{n} \ln\frac{p_1(X_i)}{p_0(X_i)} = \sum_{i=1}^{n}(1 - X_i - \ln(e-1))$$

$$L_n = n(1 - \ln(e-1)) - \sum_{i=1}^{n} X_i$$

$H_0: X_i \sim R(0,1)$, $\mu_0 = 0{,}5$ $\qquad \sigma_0^2 = \frac{1}{12}$

$$T_n = \frac{\sum X_i - \frac{n}{2}}{\sqrt{\frac{n}{12}}} \xrightarrow{d} N(0,1)$$

отвергаем H_0, если L_n слишком мал, т.е. $\sum X_i$ ~~слишк~~ слишком мал

$$\frac{\sum X_i - \frac{n}{2}}{\sqrt{\frac{n}{12}}} < z_\alpha, \quad z_\alpha - \text{ 2-квантиль}$$

$$\mu_1 = \frac{1}{e-1}\int_0^1 x e^{1-x}\,dx = \frac{e}{e-1}\int_0^1 x e^{-x}\,dx = \frac{e}{e-1}\left(-xe^{-x}\Big|_0^1 + \int_0^1 e^{-x}\,dx\right) =$$

$$= \frac{e}{e-1}\left(-\frac{1}{e} + 1 - \frac{1}{e}\right) = (1 - 2e^{-1}).$$

$\frac{e}{e-1} = \frac{e-2}{e-1} \approx 0{,}418$

$W \approx 1 - \Phi\left(z_\alpha - \frac{\mu_1 - 0{,}5}{\sqrt{\frac{1}{12n}}}\right) \xrightarrow[n\to\infty]{} 1$ критерий состоятельный

d) При $H_0: P(X_{min} < c) = 1 - (1-c)^n = \alpha$

$c = 1 - (1-\alpha)^{\frac{1}{n}}$

$W = P_{H_1}(X_{min} < c) = 1 - (1 - F_1(c))^n$ $\quad F_1(c) = \int_0^c \frac{e^{1-x}}{e-1}\,dx = \frac{e - e^{1-c}}{e-1}$

При $n \to \infty$: $c \sim \frac{\alpha}{n} \to 0$ $\quad F_1(c) \sim \frac{e}{e-1} c \sim \frac{e}{e-1} \cdot \frac{\alpha}{n}$

Тогда $(1 - F_1(c))^n \to e^{-\frac{e\alpha}{e-1}} < 1$

$W \to 1 - e^{-\frac{e\alpha}{e-1}} < 1$ не состоятельный.

Д 4.12.

H_0 – честная кость $\quad H_1$: нечестная кость с заданными вероятностями.

$H_0: p_i = \frac{1}{4}\ \forall i \quad H_1: p_1 = p_2 = \frac{1}{4},\ p_3 = \frac{1}{6},\ p_4 = \frac{1}{3}$

$\Lambda = \frac{P_{H_1}}{P_{H_0}}$

Исход	P_{H_0}	P_{H_1}
1,1	$\frac{1}{16}$	$\frac{1}{36}$
1,2	$\frac{1}{16}$	$\frac{1}{36}$
1,7	$\frac{1}{16}$	$\frac{1}{24}$
1,4	$\frac{1}{16}$	$\frac{1}{12}$
2,1	$\frac{1}{16}$	$\frac{1}{36}$
2,2	$\frac{1}{16}$	$\frac{1}{36}$
2,3	$\frac{1}{16}$	$\frac{1}{24}$
2,4	$\frac{1}{16}$	$\frac{1}{12}$
3,1	$\frac{1}{16}$	$\frac{1}{24}$
3,2	$\frac{1}{16}$	$\frac{1}{24}$
3,3	$\frac{1}{16}$	$\frac{1}{16}$
3,4	$\frac{1}{16}$	$\frac{1}{12}$
4,1	$\frac{1}{16}$	$\frac{1}{12}$
4,2	$\frac{1}{16}$	$\frac{1}{12}$
4,3	$\frac{1}{16}$	$\frac{1}{8}$
4,4	$\frac{1}{16}$	$\frac{1}{4}$

$\Lambda = 2{,}25$ $\{(3,3)\}$

$\Lambda = 1{,}5$ $\{(1,3),(2,3),(3,1),(3,2)\}$

$\Lambda = 1{,}125$

$\Lambda = 1$

$\Lambda = 0{,}75$

$\Lambda = 0{,}5625$

Для $\alpha = 0{,}2$

добавим $\{3,3\}$

$\alpha = 0{,}0625$

~~Добавим $\Lambda = 1{,}5$:~~

$\alpha = 0{,}3125$

$W = P_{H_1}((3,3)) = \frac{1}{36} \approx 0{,}0278$

Д 4.13.

$F = \frac{s_1^2}{s_2^2}$ $\quad$ При H_0: $F \sim F(n_1 - 1, n_2 - 1)$

$F = \frac{5{,}722^2}{6{,}162^2} = \frac{32{,}74}{37{,}96} \approx 0{,}862$ $\quad df_1 = 138$ $\quad df_2 = 999$

$\alpha = 0{,}05$: $\quad f_{0{,}975}(138, 999) \approx 1{,}25$ $\quad f_{0{,}025}(138, 999) \approx \frac{1}{f_{0{,}975}} \approx 0{,}8$

$0{,}862 \in [0{,}80, 1{,}25] \Rightarrow$ не отвергаем H_0.

Для ширины: $F = \frac{4{,}612^2}{5{,}055^2} \approx 0{,}832 \Rightarrow$ не отвергаем H_0

complex number
Return the imaginary part of that
std::is_pointer_v<type>
Check whether the type is a pointer type
std::is_lvalue
arithmetic type

№ Т.14.

$$z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0,1) \text{ при } H_0$$

$$\bar{x} = \frac{-1{,}11 - 6{,}10 + 2{,}42}{3} \approx -1{,}597$$

$$\bar{y} = \frac{-2{,}29 - 2{,}91}{2} \approx -2{,}6$$

$$\bar{x} - \bar{y} = 1{,}003 \quad \sqrt{\frac{2}{3} + \frac{1}{2}} \approx 1{,}080$$

$$z = \frac{1{,}003}{1{,}080} \approx 0{,}928$$

$z_{0,95} = 1{,}645 > 0{,}928 \Rightarrow$ не отвергаем H_0

$\delta = a - b \quad W(\delta) = P(z > z_\alpha \mid \delta) = 1 - \Phi\left(z_\alpha - \frac{\delta}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right)$

$\sigma_{разн} = \sqrt{\frac{2}{3} + \frac{1}{2}} = 1{,}080$

$$W(\delta) = 1 - \Phi\left(1{,}645 - \frac{\delta}{1{,}080}\right)$$

при $\delta = 0$: $W = 0{,}05$, при $\delta \to \infty$: $W \to 1$

График для Т13:

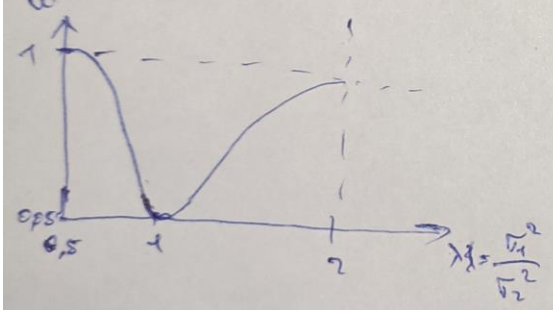
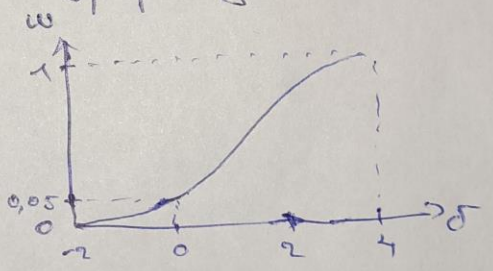
График для Т14